

SCHOOLCRAFT COUNTY AP, MI, USA

WMO: 725408

Lat: 45.975N

Lon: 86.172W

Elev: 686

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-1.2	-15.1	2.5	-2.9	-9.7	3.3	0.6	24.6	22.7	21.2	22.3	7.3	290	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	80.5	68.3	77.0	67.4	74.6	66.5	72.1	76.5	70.3	74.4	68.4	72.6	8.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	115.0	74.9	69.0	109.2	73.5	66.4	99.7	71.1	36.2	76.4	34.8	74.1	33.0	72.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.7	18.6	16.8	DB	-12.5	85.8	6.5	3.9	-17.1	88.6	-20.9	90.9	-24.5	93.1	-29.2	95.9
(5)				WB	-13.0	74.8	6.3	2.3	-17.5	76.4	-21.2	77.7	-24.8	79.0	-29.3	80.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	18.8	18.9	27.2	38.3	49.0	58.5	64.5	64.2	57.6	45.8	35.4	24.5
	DBStd	18.22	10.43	10.11	10.05	7.31	6.60	5.76	5.57	5.18	7.03	7.48	8.36	9.50
	HDD50	4429	968	871	706	353	99	6	0	0	16	177	441	792
	HDD65	8531	1433	1291	1170	800	498	206	79	78	235	595	889	1257
	CDD50	1514	0	0	1	3	67	261	448	439	245	48	2	0
	CDD65	140	0	0	0	0	1	11	62	52	14	0	0	0
	CDH74	654	0	0	0	0	8	61	344	217	24	0	0	0
	CDH80	78	0	0	0	0	1	7	52	18	0	0	0	0
Wind	WSAvg	7.7	8.2	8.3	8.4	8.7	8.0	6.7	6.5	6.4	7.0	7.7	8.0	8.0
Precipitation	PrecAvg	30.7	1.7	1.2	2.0	2.5	3.1	3.1	3.0	3.0	3.5	3.2	2.5	1.9
	PrecMax	39.5	5.7	3.2	4.9	4.8	5.6	6.8	6.7	4.4	8.1	5.8	4.8	3.9
	PrecMin	24.4	0.7	0.2	0.3	0.8	1.6	1.0	0.6	0.9	1.1	1.2	0.9	0.2
	PrecStd	3.8	1.1	0.7	1.3	1.0	1.0	1.4	1.6	0.8	1.4	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	43.2	55.1	63.0	73.2	81.0	84.5	81.9	77.3	67.6	56.9	47.7
		MCWB	37.7	41.7	47.5	48.6	58.5	65.8	69.0	69.4	67.3	62.2	51.0	46.5
	2%	DB	37.3	37.3	48.0	57.1	68.2	74.8	81.1	79.1	73.2	64.1	53.0	42.9
		MCWB	35.3	35.6	42.8	45.4	57.4	63.6	68.7	68.8	66.7	59.5	49.9	40.7
	5%	DB	35.9	35.9	43.4	53.9	64.0	72.1	78.6	76.0	71.7	62.2	49.8	39.4
		MCWB	34.2	34.3	38.7	43.8	54.8	62.2	68.0	67.7	65.7	57.7	47.0	37.2
	10%	DB	33.7	33.7	39.1	49.9	61.0	69.6	74.9	73.6	69.6	58.8	47.5	37.1
		MCWB	32.4	31.8	35.8	42.7	52.7	61.5	66.8	66.3	64.5	54.8	45.3	35.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	42.2	50.7	52.1	64.0	69.9	74.4	74.4	70.8	64.0	53.3	46.3
		MCDB	40.2	44.4	53.9	57.4	67.6	74.9	79.9	78.8	74.0	65.4	55.3	46.8
	2%	WB	36.3	36.7	43.8	48.4	60.0	67.0	72.3	72.2	68.5	60.9	50.3	41.1
		MCDB	37.6	37.5	47.3	53.9	65.0	72.0	77.3	76.3	71.7	63.4	52.4	43.0
	5%	WB	34.0	33.9	39.6	46.2	56.9	64.6	70.4	70.6	66.9	58.3	47.3	38.2
		MCDB	35.3	35.1	43.1	50.7	61.8	69.0	75.2	74.3	70.0	61.5	48.8	39.7
	10%	WB	32.0	31.4	36.8	44.0	54.1	62.8	68.4	68.6	65.2	55.4	45.5	35.9
		MCDB	33.2	33.0	39.2	49.0	58.8	66.6	73.2	72.4	67.8	58.1	46.7	37.3
Mean Daily Temperature Range	5% DB	MDBR	12.9	14.6	15.0	15.9	17.8	18.1	18.5	18.1	17.0	14.8	12.1	12.0
		MCDBR	10.9	12.0	15.8	22.1	22.8	21.6	21.3	19.5	17.2	17.1	14.6	12.0
		MCWBR	10.2	10.5	11.6	13.3	13.7	12.8	12.3	12.4	11.9	13.1	12.2	11.5
	5% WB	MCDBR	10.5	11.7	14.6	18.4	17.8	17.2	17.7	16.5	14.2	14.6	13.5	11.7
		MCWBR	10.2	10.8	11.4	13.1	13.0	11.9	12.1	12.1	11.0	12.4	12.0	11.5
Clear-Sky Solar Irradiance	taub	0.268	0.281	0.309	0.353	0.381	0.395	0.391	0.385	0.366	0.333	0.302	0.270	
	taud	2.403	2.347	2.383	2.367	2.353	2.345	2.371	2.389	2.440	2.513	2.518	2.452	
	Ebn at Noon	268	286	292	287	281	277	276	274	270	265	254	256	
	Edh at Noon	22	29	32	36	38	39	37	35	31	25	20	19	
All-Sky Solar Radiation	RadAvg	374	664	1091	1440	1766	1907	1943	1646	1238	735	399	287	
	RadStd	35	78	107	146	138	144	89	107	91	83	47	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page